The D. G. Ruparel College
OF ARTS, SCIENCE AND COMMERCE
Department of Computer Science • Information Technology

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TITLE :

Aim:- To Implement basic Publisher-Subscriber using MQTT. Publish messages using paho mqtt. Subscribe and control LED.

* Hardware Requirements
1. Raspberry Pi Model A/B/B+
2. Jumper wires (Female to Male)

* Software Requirements.
1. Raspbian Stretch OS
2. MQTT Dashboard (From. Google Playstore)

1. Connect your 4 digit 7 segment display with Respberry Pi's GPIO pins.

LED Pin	Function	RPI Physical Pin	Raspberry Function
VCC	POWER	12	GPIO 18
GND	GND	6	GND

*Code :-

```
import paho.mqtt.client as mqtt
import RPi.GPIO as GPIO

# ----- GPIO SETUP ----
GPIO.setmode(GPIO.BOARD)
GPIO.setwarnings(False)

# Physical Pin 12
LED = 12

GPIO.setup(LED, GPIO.OUT, initial=GPIO.LOW)

#---- MQTT CONNECT ----
def on_connect(client, userdata, flags, reason_code,
               properties):
  print("Connected with result code:", reason_
                                       code)

  #IMPORTANT: same topic as your mobile app
  client.subscribe("led")

  print("Subscribed to topic : led")

#----- MQTT MESSAGE ----
def on_message(client, userdata, msg):
```

```
# Convert received message to lowercase
message = msg.payload.decode("utf-8").
                    strip().lower()

print("Received:", message)

# LED @1 ON
if message == "on":
    GPIO.output(LED, GPIO.HIGH)
    print("LED ON")

# LED OFF
elif message == "off":
    GPIO.output(LED, GPIO.LOW)
    print("LED OFF")

# Unknown message
else:
    print("Unknown message:", message)

# ----- MQTT CLIENT ----
client = mqtt.Client(
    callback_api_version=mqtt.CallbackAPIVersion.
                        VERSION2
)
```

```
client.on_connect = on_connect
client.on_message = on_message

#Connect to MQTT broker
client.connect("broker.emqx.io", 1883, 60)

print("Waiting for MQTT messages....")

#keep program running
client.loop_forever()
```